

Scott J. Gilbert

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Aspiring computer engineer with strong foundations in hardware design, embedded systems, and full-stack development. Skilled in C/C++, Python, systems engineering, low-level programming, and data processing, with hands-on experience in real-time systems, machine learning, and DevOps.

Education

University of Illinois Urbana-Champaign

GPA: 4.00/4.00

B.S., Computer Engineering, August 2025 - May 2029

The Grainger College of Engineering

Activities and Societies: Campus Honors Program, Illini Solar Car, Project: Code, Illini Redstone Computing, Engineers without Borders, Marching Illini
Relevant Coursework: Differential Equations (MATH 285), Intro to Computing (ECE 120), Computational Linear Algebra (MATH 257), Intro to Electronics (ECE 110), Computers & Programming (ECE 220), Quantum Physics (PHYS 214), Thermodynamics (PHYS 213), Data Structures (CS 225)

James B. Conant High School

High School Diploma, August 2021 - May 2025

Weighted GPA: 4.92/4.00 | **Unweighted GPA:** 4.00/4.00

Activities: Congressional Debate, Marching Band, Jazz Band, Honor Band/Orchestra, Cross Country, Track and Field, National Honor Society

Relevant Coursework: Multivariable Calculus (Calculus III), AP Calculus BC, AP Computer Science Principles, AP Computer Science A, AP Physics C

Professional Experience

Computer Administrator

October 2025 - Present

University of Illinois Campus Honors Program (CHP)

Urbana, Illinois

- Maintain CHP's computing hardware infrastructure, external storage integrations, and web services across multiple internal/external providers.
- Collaborate with CHP staff, IT services, and other teams to modernize computing/internet resources and increase their accessibility to students.
- Supported data synchronization between university admissions and CHP databases, reducing entry time for record processing by up to two weeks.

Full-Stack/AI Intern

June 2025 - August 2025

Team2Go Inc.

Yongin-si, Gyeonggi, South Korea (Remote)

- Engineered Dockerized systems for internal and external-facing tools, enhancing integration between backend applications and user interfaces.
- Integrated custom OpenAI GPT models with streaming output, document-based vector embedding, and rigid boundary/instruction declaration.
- Collaborated with executives through regular progress updates and delivered projects ahead of schedule while exploring LLM design/psychology.

Leadership/Involvement

Array/MPPT Lead & Webmaster

Illini Solar Car | Oct 2025 - Present

- Develop maximum power point tracking firmware, use OpenGL-based tools to determine solar array layout specifications, and manage budgets/sponsorship agreements in the tens of thousands of dollars, contributing to overall electrical and vehicle optimization.
- Serve on the executive board as Webmaster, integrating software across hosting services and managing containerization, DNS, and proxies.

President & Systems Administrator

Illini Redstone Computing | Jan 2026 - Present

- Manage day-to-day financial, administrative, and technical decision-making while acting as primary liaison to the university/other organizations.
- Develop and maintain a network of containerized applications across gaming, authentication, user management, and data pipelining services.

Eagle Scout

Scouting America | Mar 2018 - Dec 2024

- Planned and organized a coordinated nature restoration effort where volunteers contributed over 100 working hours.
- Collaborated with sponsors to secure necessary resources/tooling to finish the project a week ahead of schedule with zero workplace incidents.

Projects

Delos (Solar Electric Vehicle)

October 2025 - Present

- Modernized firmware to resolve five-year-old systemic issues, implement boost converter power delivery, and CAN for reliable messaging.
- Helped design/build ISC's most efficient array (1.5kW, 6 m²) and no-failure wheel, dash, and MPPT subsystems, powering the car for 650+ miles.
- Contributed to development and installation of the car's electrical/computer systems, including enclosures, cooling, wiring, and ergonomics.
- Fabricated carbon-fiber shells/enclosures, conducted multi-stage battery testing, and conducted full integration of all car electrical components.

Personal Content System

May 2025 - Present

- Developing flexible microservices-based architectures integrating a portfolio, blog, authentication, account management, AI-driven content development, workflow automation & CI/CD tooling and service integration, and a rich, data-driven backend through container orchestration.
- Modernizing backend with scalable technologies like GraphQL for improving performance and expanding application and cross-platform use cases.

Illinois Congressional Debate Association Website

May 2024 - August 2025

- Led full-stack application development, improving front-end utility with React and performance/scalability with a backend service and CMS.
- Programmed, tested, and deployed features improving mobile accessibility, information flow, site security, and page/component load times.
- Achieved strong search visibility, reaching ~20K impressions/6% CTR over four months, ranking among Google's top ten student congress sites.

Skills

- **Programming:** C, C++, Python, SQL, Java, Assembly, TypeScript, PHP, Wolfram
- **Datastores:** MySQL, SQLite, MariaDB, PostgreSQL, pgvector, S3, Uploadthing, Backblaze
- **Numerical Analysis:** Jupyter Notebook, NumPy, SciPy, Matplotlib, Mathematica, Excel, Google Sheets
- **Embedded Systems/Microcontrollers:** Arduino, ELEGOO, KiCad, NXP, ARM Cortex, AMD Vivado
- **Industry Tools:** Docker, Podman, MCUXpresso, LTSpice, Git, GitHub, Visual Studio Code, JetBrains, Google Workspace, Microsoft Office, Slack
- **Workflow Automation & CI/CD:** Vercel, Claude Code, GitHub Copilot, GitHub Actions, n8n, Make
- **Technical Expertise:** Embedded Systems, Firmware Design, Backend Development, PCB Design, AI/ML Application, Virtualization/Containerization

Awards: Dean's List, National Merit Scholarship, AP Scholar with Distinction, Illinois State Seal of Biliteracy, Gold President's Volunteer Service Award